Arnab Dey

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Professional Summary

Cloud & Platform Engineer with **15+ years in IT** and **9+ years building and operating Azure platforms** that real product teams depend on — most recently a **GitOps self-service Azure Databricks platform at Volvo Cars** (PR-driven Terraform via GitHub Actions, lead time days → same-day) and a **~100-subscription Azure CAF Enterprise Scale governance estate at Stena Metall** delivered through the production Terraform + Azure DevOps pipeline. SRE-aligned by practice: KQL-driven monitoring at Stena that contributed to a **~20% reduction in downtime**, 24×7 incident handling on customer-critical Linux estates at HP, and observability stacks (ELK, Prometheus/Grafana, Azure Monitor) built for production. Deep on Azure compute / networking / identity / storage, Terraform module-library patterns, **GitHub Actions and Azure DevOps**, and security-first delivery (custom RBAC, Key Vault compliance, SP rotation design). Stockholm-based, fluent in English.

Core Skills

Microsoft Azure (primary) — CAF Enterprise Scale / Landing Zones (~100-subscription estate), **Compute** (VM/VMSS, App Services), **Networking** (VNet, NSG, Private Endpoints), **Identity** (Entra ID, RBAC including custom role definitions, Service Principals), **Storage** (Blob, ADLS, disk-encryption-sets), Key Vault, AKS, ACR, Azure Monitor, Log Analytics / KQL, Azure Resource Graph, Azure Policy.

CI/CD — GitHub Actions and Azure DevOps (both) — **GitHub Actions** with reusable workflows for Volvo Cars Terraform delivery (`tf-validate` on PR, `tf-deploy` on merge); **Azure DevOps Pipelines** for Stena governance Terraform and Greenstone .NET microservices; reusable templates, automated quality gates (`fmt/validate/plan` separated from `apply`).

Infrastructure as Code — Terraform (AzureRM, AWS, OpenStack, Databricks providers): reusable module libraries with opinionated, governance-friendly defaults; archetype-driven CAF deployment; remote state in Azure Blob with backend config injected from CI environment secrets; module versioning, drift handling, state inspection. **ARM templates** (closest analogue to Bicep — same resource model, transpilation target — comfortable picking up Bicep on the job).

Site Reliability practices — Observability (Azure Monitor, Log Analytics / KQL, Prometheus, Grafana, ELK Stack); **KQL-driven monitoring contributed to ~20% downtime reduction at Stena**; production ELK stack with X-Pack security and snapshot repositories at Volvo; 24×7 incident handling at HP across customer-critical Unix/Linux estates; reliability-by-design via autoscale + auto-termination defaults baked into Terraform modules.

DevSecOps & compliance — **Custom Azure RBAC role definitions** (separation-of-duties for ransomware resilience); **tenant-wide Key Vault compliance audit** (soft-delete, purge protection, RBAC authorisation); **Service Principal lifecycle + credential-rotation design** (Logic Apps + Graph

`addPassword`); SAML 2.0 federation replacing long-lived IAM keys; Azure Backup compliance reporting; cost-allocation tagging across landing zones.

Backend language familiarity — **C# / .NET**: Azure DevOps pipelines for .NET microservices at Greenstone Financial Services (Docker Swarm + ACR + Azure Monitor); PowerShell **App Service .NET runtime enumeration** across the Stena CAF estate.

Kubernetes & containers — AKS, EKS, **Rancher RKE**, K3s; Docker, Helm, Kubeflow, ArgoCD; container registries (ACR, ECR, Nexus).

Cloud (other) — AWS (EC2, S3, VPC, EKS, IAM, ECR, ALB, KMS); OpenStack (Deutsche Telekom 5G commercial pilot).

Scripting — Python (REST/SCM automation, CLI tooling), PowerShell (Azure governance — SP lifecycle, ACR token expiry, runtime audits, budget automation), Bash.

Languages — English (fluent), Swedish (SFI C-level).

Professional Experience

DevOps Engineer — AFRY Digital Solutions AB

May 2025 – Present | Stockholm, Sweden

Assignment: Ericsson AB — CI Engineer, Tiger Test Tools (Sep 2025 – Present)

- Maintain and support the end-to-end **CI flow for radio-communication smoke and regression testing** — Git, Gerrit, Jenkins, Docker, Linux. Close the loop between pipeline failures and physical lab state when test rigs misbehave.

Platform engineering (internal reference project — public on GitHub)

- Designed and operate a **GitOps Kubernetes platform**: K3s + ArgoCD + Prometheus + Grafana + cert-manager + sealed-secrets. Live demonstration of declarative platform foundations.

Cloud Engineer — Stena Metall AB

Aug 2024 – Jan 2025 | Gothenburg, Sweden

Cloud Infrastructure Engineer on the central Cloud Office (Governance) of a mature **Azure CAF Enterprise Scale** estate — **~100 subscriptions, multi-management-group hierarchy**, production workloads for a multi-billion-SEK industrial group. All infrastructure delivered as **Terraform via Azure DevOps pipelines**.

- **Authored two custom Azure RBAC role definitions** — a "Backup Remover" role and a Contributor variant with explicit `notActions` — to enforce **separation-of-duties** between resource management and backup deletion (a ransomware-resilience control missing from Azure's built-in Contributor). **Wired into the CAF archetype framework** and shipped through the production Terraform + Azure DevOps pipeline. **38 commits across the core-governance repo.**

- **Tenant-wide Key Vault compliance audit** — KQL against Azure Resource Graph for soft-delete retention, purge protection, and RBAC authorisation; CSV inventories and a documented review for architecture leadership.
- Built **Service Principal lifecycle tooling** — KQL audit + PowerShell expiring-credential inventory + a Logic Apps / Graph `addPassword` rotation design.
- Implemented **Azure Backup policy compliance reporting** with per-VM exempt-rationale tracking.
- PowerShell tooling: subscription budget automation, ACR token expiry inventory, .NET runtime enumeration across App Services, Hybrid Benefit enablement, cost-allocation tagging across landing zones.
- **KQL-based monitoring contributed to a ~20% reduction in downtime** across supported services.
- Published sanitised sister repos on GitHub: `iac-azure-core-governance` (CAF Enterprise Scale governance), reusable Terraform module library, and Azure cost-optimisation templates.

Senior Software Engineer (DevOps) — Capgemini Sverige AB

Sep 2021 – Jul 2024 | Gothenburg, Sweden

Assignment: Volvo Cars — Car Safety R&D (Abakus MLOps / AI Platform)

Self-service compute platform via GitOps — supporting product (data science) teams

- Designed and owned a **GitOps self-service pattern for Azure Databricks**: data scientists request compute via a GitHub issue → YAML manifest PR → **GitHub Actions runs `terraform fmt/validate/plan`** for review → merge triggers `terraform apply`.
- Authored the **Terraform module** with opinionated, governance-friendly defaults baked in: single-user security mode, **30-minute auto-termination, autoscale 1→3 workers**, cluster-log shipping to DBFS, `CAN_RESTART` permissions — **reliable and cost-efficient by construction**.
- Remote state in **Azure Blob Storage** with backend config injected at init from GitHub environment secrets; two-workflow contract (`tf-validate` on PR, `tf-deploy` on merge).
- **Outcome:** provisioning lead time fell from days to same-day; every cluster traceable to a PR, manifest, and named owner; long-tail idle-cluster spend eliminated.

Kubernetes platform modernisation

- Contributed to modernising a ~36-node CAE/HPC cluster: OS migration SLES → Ubuntu LTS; stand-up of **Rancher-managed Kubernetes (RKE)**; **Kubeflow** via Helm.
- Built a **production-grade ELK stack** (Elasticsearch, Logstash, Kibana) on LXD with X-Pack security, SSL, and snapshot repositories for observability.

Dual-cloud benchmarking

- Built **two matched Terraform stacks** (AWS: EC2 + S3 + KMS + IAM | Azure: VM + Premium BlockBlobStorage) as identically-shaped, disposable test rigs for object-storage benchmarking.

Senior Business Consultant (DevOps) — Tech Mahindra

Feb 2017 – Sep 2021 | Bangalore, India | Sydney, Australia

Assignment: Greenstone Financial Services — Insurance Portal

- Led DevOps transformation for **.NET / C# microservices**: **Azure DevOps pipelines**, **Terraform for Azure infrastructure**, Docker + Docker Swarm with **Azure Container Registry**, Azure Monitor for observability.

Assignment: Cisco Systems — CD 2.0 Continuous-Delivery Platform on AWS

- **Authored the Terraform** for Cisco's CD 2.0 platform on AWS — VPC, subnets, security groups, **EKS cluster, IAM roles, Jenkins on EKS, ECR integration, build-agent fleet, ALB, DNS** — extending an internal reference architecture.
- Wrote a separate **Terraform stack for an AWS observability environment** (EKS + Prometheus Operator via Helm + Grafana + persistent storage + ingress).
- Closed out Cisco's long-running SVN-to-Git migration via Python automation against the Bitbucket REST API — with retry logic and batching for SCM rate limits.

Assignment: Deutsche Telekom — 5G BSS Slicing Pilot on OpenStack

- Owned the **DevOps platform side** of DT's 5G Network Slicing commercial pilot, delivering Blue Marble's ~30-microservice BSS suite.
- **Terraform against the OpenStack provider** provisioning the Jenkins / Nexus / Tuleap / Rancher / Keycloak / ELK estate.
- **~46 Jenkins jobs** building BMP microservices, tagging build number + short git SHA, pushing to internal Nexus, updating Kubernetes manifests, triggering Rancher rolling updates.
- Owned a **cross-tenant migration weekend** rehoming 54 Tuleap repositories, Jenkins jobs, Nexus artifacts, Keycloak realm, and live Kubernetes workloads via Python/shell automation.

Senior AEM Administrator (Cloud) — ICF Next

Sep 2015 – Jan 2017 | Bangalore, India

Assignment: Channel 7 (Australia) — AEM web platform on AWS

- Built and operated a production AEM web platform on AWS for a national broadcaster — VPC, EC2 lifecycle, Apache HTTPD + Dispatcher hardening, SSL termination, **SAML 2.0 federation (Azure AD / ADFS → AWS IAM)** replacing long-lived IAM user keys.
- Authored **Ansible roles** for AEM install, Apache/Dispatcher deployment, SSL, and base OS — turning instance rebuilds into a runbook rather than a discovery exercise.

Unix / Linux Systems Administrator — HP Enterprise Services

Nov 2011 – Aug 2015 | Bangalore, India

Three-and-a-half years on HP's Global Operations Center under Best Shore Services, rotating across enterprise outsourcing customers (HUL, E.ON, Con-way Inc.) — **24×7 incident handling on customer-critical Linux/Unix estates** (140+ servers), LVM/SAN, kernel tuning, SOP authorship for production runbooks, HP Service Manager / OpenView / Server Automation.

Education

B.Tech in Computer Science — West Bengal University of Technology, Kolkata, India (2005 – 2009)

Certifications

- **Microsoft Certified: Azure Fundamentals (AZ-900)** — Mar 2023 *(Credential ID: 801D970BAA49297)*
- **AWS Certified Solutions Architect – Associate (SAA-C03)** — *in progress*